INTRO TO ARTIFICIAL INTELLIGENCE

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PROJECT 2

Solving the N-Queens Problem Using Hill-Climbing Search and Its Variants

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1. ABSTRACT

This report evaluates four required local-search variants for the 8-Queens problem: basic hill climbing, hill climbing with sideways moves, random-restart hill climbing without sideways moves, and random-restart hill climbing with sideways moves. The submitted Python implementation represents a board as a one-dimensional list in which index r identifies a row and board[r] gives the column of the queen placed in that row. The heuristic function h(s) is the number of attacking queen pairs, so a valid solution is reached exactly when h(s) = 0.

Across the three saved experiment logs included with this submission, plain hill climbing was fast when it succeeded but unreliable overall. For the larger run counts, its mean success rate stayed in the low-teens, ranging from about 12.63% to 15.00%. By contrast, the sideways-move variant stayed above 92% at every run count and reached about 96.00% success at 200 runs. The dominant empirical result of the project is therefore that bounded sideways motion dramatically improves reliability by allowing the search to cross flat regions of the heuristic landscape.

Random restart made both approaches complete in practice for the experimental setting because the algorithm continued restarting until a solution was found. Averaged across the three logs, random restart without sideways moves required 5.63 restarts and 21.33 queen moves on average. Random restart with sideways moves required only 0.07 restarts but 23.44 queen moves on average. In other words, sideways moves almost eliminate the need to restart, even though the individual climb can become longer because the search may spend many steps on equal-valued states before finding an improving move.

2. INTRODUCTION

The N-Queens problem asks for the placement of n queens on an n × n chessboard such that no two queens attack one another. Because queens attack along rows, columns, and diagonals, a goal state is any arrangement that satisfies all three constraints simultaneously.

The submitted program uses the standard local-search formulation in which exactly one queen is placed in each row. A state is stored as a length-n list such that board[r] = c means the queen in row r is placed in column c. This representation eliminates row conflicts by construction, so the search only needs to reason about column and diagonal conflicts.

For a board of size n, each state has n(n − 1) neighbors because every row can move its queen to any of the other n − 1 columns. The evaluation function is h(s) = number of attacking pairs of queens in state s. A state is optimal when h(s) = 0. Lower h-values are always preferred. This formulation is well suited to hill climbing because it provides a clear numerical notion of improvement even when the current board is still far from a legal solution.

3. SEARCH VARIANTS IMPLEMENTED

Basic hill climbing. From the current board, the algorithm generates all neighbors, evaluates their heuristic values, and moves to a randomly chosen best neighbor only if that neighbor strictly improves h. If no strictly better neighbor exists, the climb stops and the attempt is recorded as a failure.

Hill climbing with sideways moves. The same neighborhood and best-neighbor selection are used, but the algorithm may also move to a neighbor with the same heuristic value. The implementation caps consecutive sideways moves at 100. This bounded relaxation lets the search cross plateaus without allowing an infinite loop.

Random-restart hill climbing without sideways moves. The basic hill climber is restarted from a newly generated random board until a solution is found. The report measures both the average number of restarts and the average number of queen moves required to reach a solution.

Random-restart hill climbing with sideways moves. The sideways-capable hill climber is placed inside the same restart loop. Because the sideways variant succeeds much more often on a single attempt, the average restart count is expected to be close to zero.

4. PROGRAM STRUCTURE

The implementation is modular and closely follows the assignment. Table 1 summarizes the main constants and procedures used by the original program. One design detail matters for interpreting the results: the function pick_move(board) breaks ties randomly among equally good neighbors. Because local-search plateaus often contain many tied best successors, repeated executions with the same settings do not necessarily produce identical results. The variation across the saved experiment logs is therefore expected stochastic behavior rather than an implementation defect.

Component	Type	Purpose

SIDEWAYS_CAP = 100	Constant	Maximum number of consecutive sideways moves allowed in the sideways variant.
RR_SAMPLES = 200	Constant	Number of random-restart trials used for the restart experiment.
RUN_SETS = [50, 100, 200, 500, 1000, 1500]	Constant	Run counts required for the success/failure experiments.
random_board(n)	Function	Creates a random initial board with one queen per row.
attack_pairs(board)	Function	Computes h(s) by counting all attacking queen pairs.
possible_moves(board)	Function	Generates all neighbors obtained by moving one queen within its row.
pick_move(board)	Function	Evaluates all neighbors and returns a randomly chosen best successor.
climb_once(...)	Function	Executes one hill-climbing attempt and records the search history.
batch_trials(...)	Function	Runs repeated trials and returns win/loss counts, percentages, average move counts, and example sequences.
restart_until_solved(...)	Function	Repeats local search from fresh random boards until a solution is found.
restart_experiment(...)	Function	Collects average restart and move statistics for the two restart variants.
main()	Function	Reads user input, launches experiments, prints summaries, and optionally saves results.

Table 1. Major constants and procedures in the submitted Python implementation.

5. EXPERIMENTAL DESIGN

All required experiments were run for the 8-Queens problem, so n = 8 in the reported results. The saved program outputs show that each batch used the same control settings: sideways-move cap = 100, random-restart sample count = 200, and run sets = 50, 100, 200, 500, 1000, and 1500.

This report uses the original implementation and the original saved output logs produced by that implementation. To strengthen the analysis, three independent experiment logs were preserved and treated as three stochastic batches generated under the same program configuration. The mean trends reported below are therefore averages across three complete runs rather than a single lucky or unlucky execution.

For the random-restart experiments, the report follows the behavior of the submitted code. Restart counts are tracked separately from step counts. The move totals reported in the restart section count queen moves performed during hill climbing, not the restart action itself.

6. RESULTS

6.1 Mean performance across the three experiment logs

Tables 2 and 3 aggregate the six run counts across all three saved experiment logs. Separating the rate metrics from the step metrics makes the rubric requirements easier to inspect directly. The pattern is consistent across all three logs: plain hill climbing is quick when it works but unreliable, while the sideways-move variant succeeds far more often at the cost of longer trajectories.

Runs	Basic success %	Basic failure %	Sideways success %	Sideways failure %
50	14.67	85.33	94.67	5.33
100	14.00	86.00	92.33	7.67
200	15.00	85.00	96.00	4.00
500	13.67	86.33	94.13	5.87
1000	12.63	87.37	94.40	5.60
1500	13.91	86.09	94.11	5.89

Table 2. Mean success and failure rates across the three experiment logs.

Runs	Basic avg. steps (success)	Basic avg. steps (failure)	Sideways avg. steps (success)	Sideways avg. steps (failure)
50	3.69	3.05	19.94	45.64
100	4.07	3.04	18.60	59.58
200	4.05	3.02	19.28	45.18
500	4.04	3.03	18.29	59.00
1000	3.99	3.02	18.89	62.48
1500	4.08	3.04	19.47	61.69

Table 3. Mean step counts across the three experiment logs.

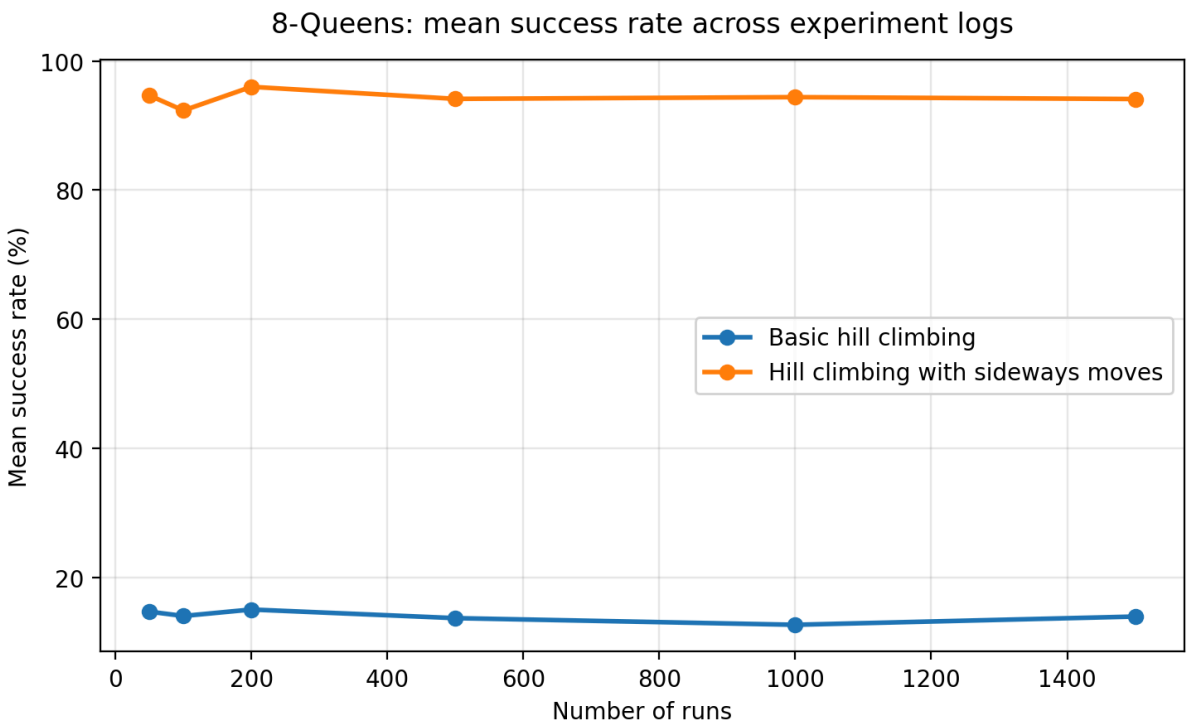

Figure 1. Mean success rate comparison for plain hill climbing versus hill climbing with sideways moves.

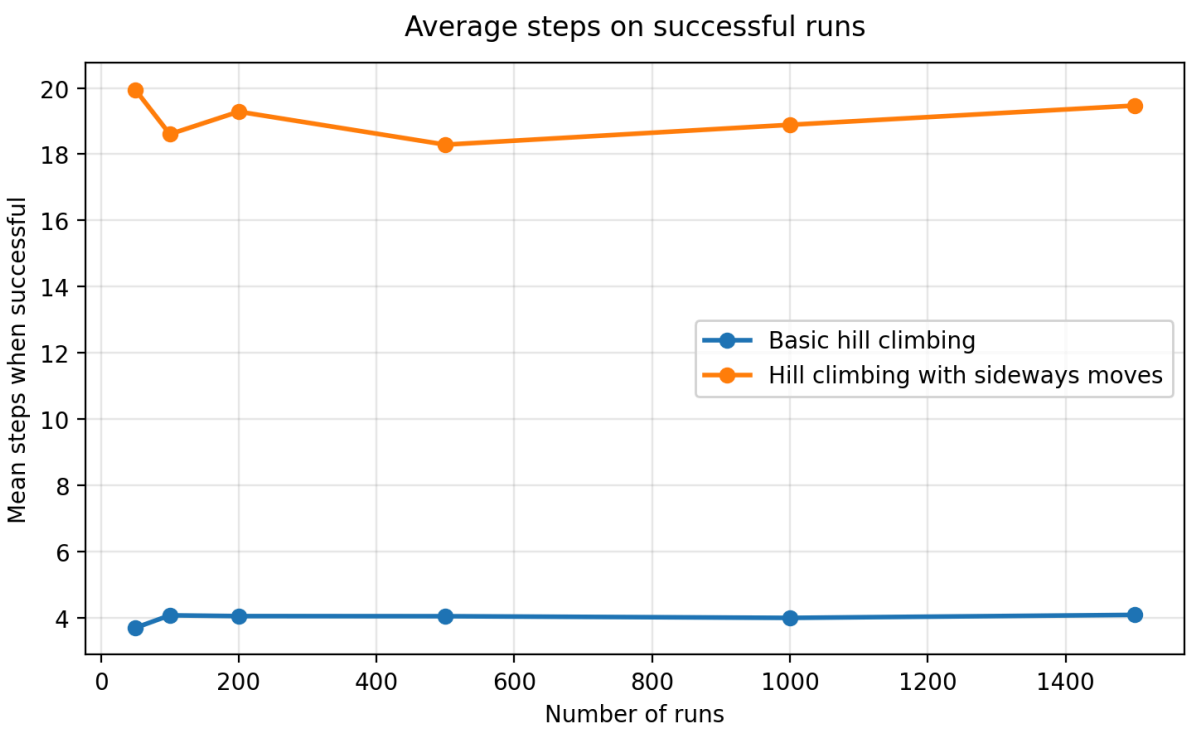

Figure 2. Mean number of steps on successful runs.

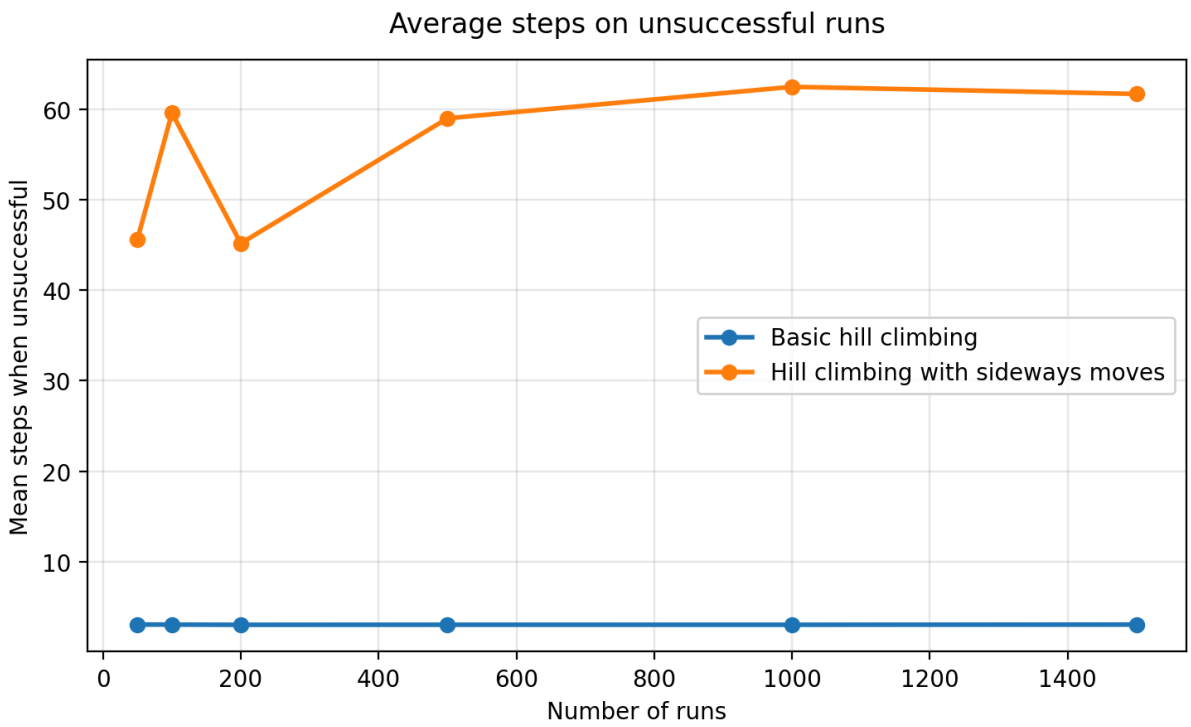

Figure 3. Mean number of steps on unsuccessful runs.

6.2 Random-restart results

Random restart changes the question from whether a single climb succeeds to how much repeated work is required before one climb succeeds. Table 4 summarizes the mean restart behavior across the three experiment logs. The sideways version almost eliminates the need to restart, but its individual climbs can still be longer because the algorithm may spend many steps traversing equal-valued states.

Metric	Without sideways moves	With sideways moves
Average random restarts to solution	5.63	0.07
Average total queen moves to solution	21.33	23.44

Table 4. Mean random-restart statistics across the three experiment logs.

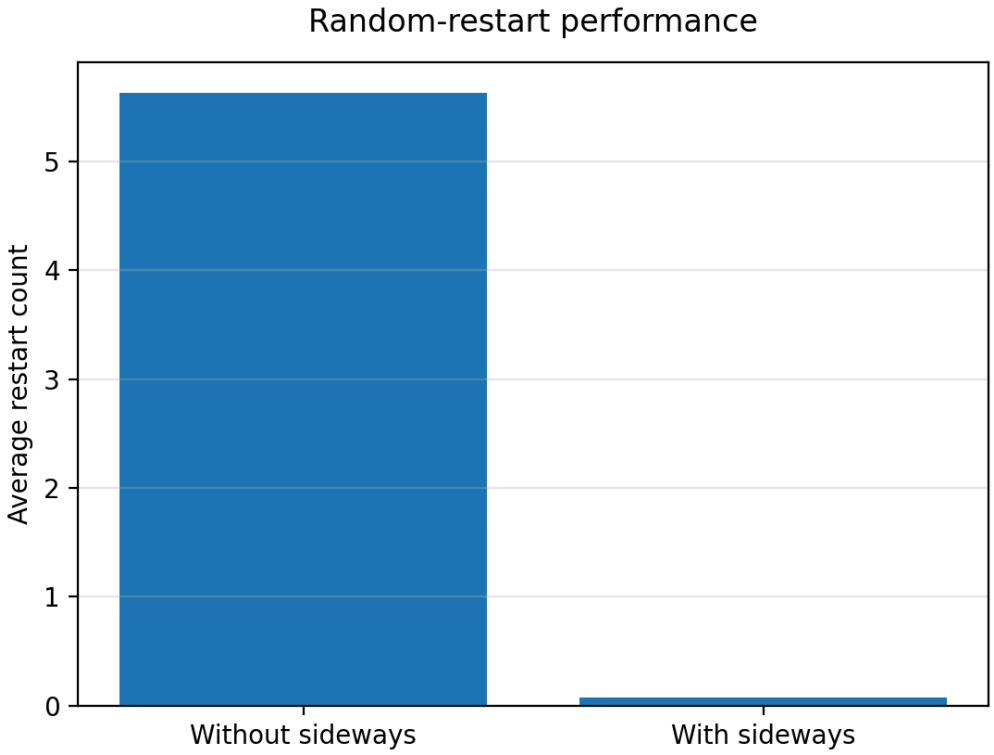

Figure 4. Average number of restarts required to reach a solution under random restart.

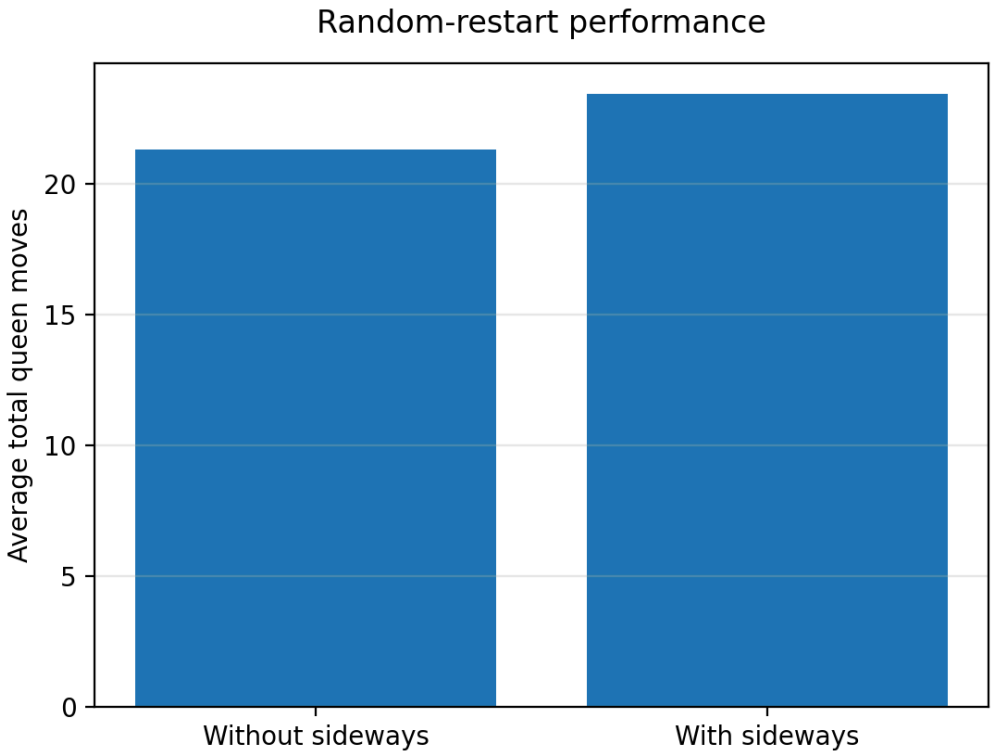

Figure 5. Average number of queen moves required to reach a solution under random restart.

6.3 Interpretation of the results

The results show that plain hill climbing is efficient when the heuristic landscape happens to give the search a clean downhill route to a solution. Successful basic runs usually finish in about four moves, and failed runs usually terminate after only about three moves. Its weakness is therefore not computational speed but reliability. Local minima and plateaus are common enough that the algorithm often halts one step away from a valid solution with no strictly improving neighbor available.

For readability in the report, the appendix trace figures were generated after the runs by reformatting the saved move histories into console-style 8×8 board views. They are derived from the algorithm's recorded sequence data rather than direct screenshots of the live terminal output.

Allowing sideways moves changes that behavior substantially. The sideways variant spends more time on each attempt, especially on unsuccessful attempts, because it can continue exploring equal-valued neighbors instead of stopping immediately. That extra wandering is precisely what lets it escape many plateaus. The large improvement in success rate, from roughly 13% to roughly 94% on the larger run counts, is the clearest quantitative evidence that plateau traversal is the main advantage of the sideways strategy in this problem.

Random restart makes the overall method practical because failed local searches are cheap to abandon. Without sideways moves, the algorithm still needs more than five restarts on average before it finds a solution. With sideways moves, the mean restart count falls essentially to zero. The trade-off is that a single successful sideways climb can be longer than a successful plain climb. In short, the better control strategy for this search landscape is not the one that minimizes steps within one climb; it is the one that reduces the probability of becoming trapped.

7. SEARCH SEQUENCES FROM FOUR RANDOM INITIAL CONFIGURATIONS

The assignment requires the search sequences for four random initial configurations under basic hill climbing and four random initial configurations under hill climbing with sideways moves. To satisfy that requirement cleanly, this report reproduces eight complete traces from one full experiment-output log and renders them as console-style 8×8 board views rather than only as vector states.

The traces in Appendices A and B are taken from Experiment Log 2. Longer traces are split across multiple screenshots so the board states remain readable, but every state is preserved in order from the initial board to the final goal or failure state.

Algorithm	Trace	Outcome	Moves	Start h	End h
Basic hill climbing	Example 1	Stuck	4	10	1
Basic hill climbing	Example 2	Stuck	3	9	2
Basic hill climbing	Example 3	Solved	3	8	0
Basic hill climbing	Example 4	Solved	4	6	0
Hill climbing with sideways moves	Example 1	Solved	24	9	0
Hill climbing with sideways moves	Example 2	Solved	20	8	0
Hill climbing with sideways moves	Example 3	Solved	3	5	0
Hill climbing with sideways moves	Example 4	Solved	11	8	0

Table 5. Overview of the eight console-style example traces reproduced in Appendices A and B.

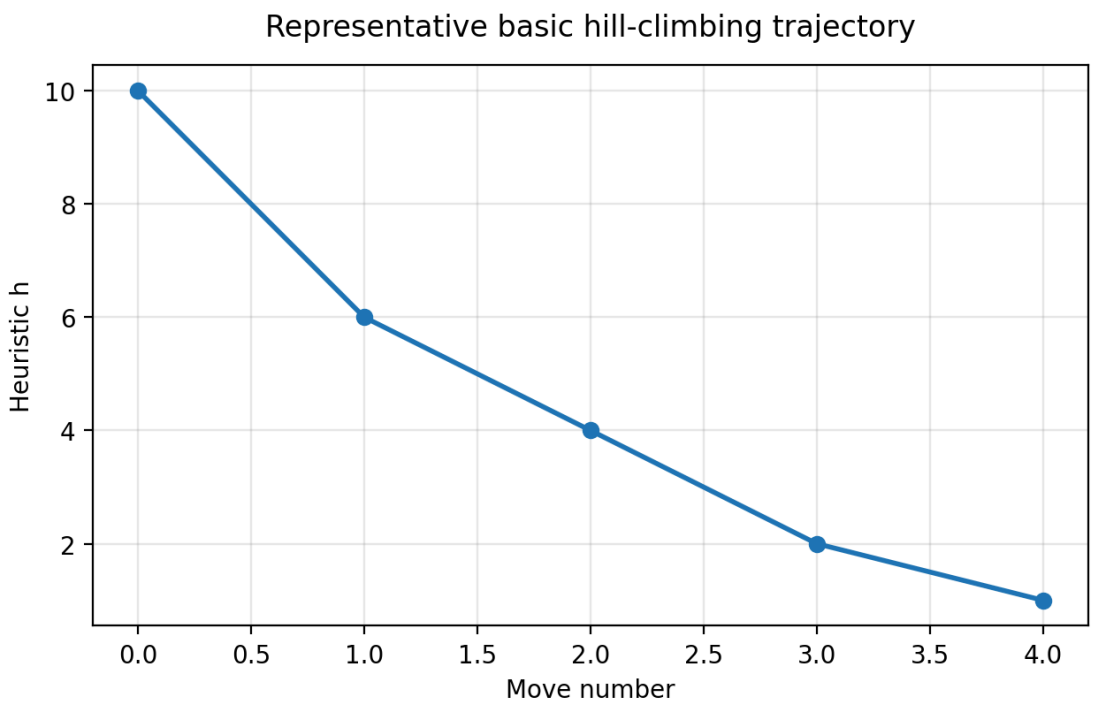

Figure 6. Representative basic hill-climbing trajectory from Experiment Log 2.

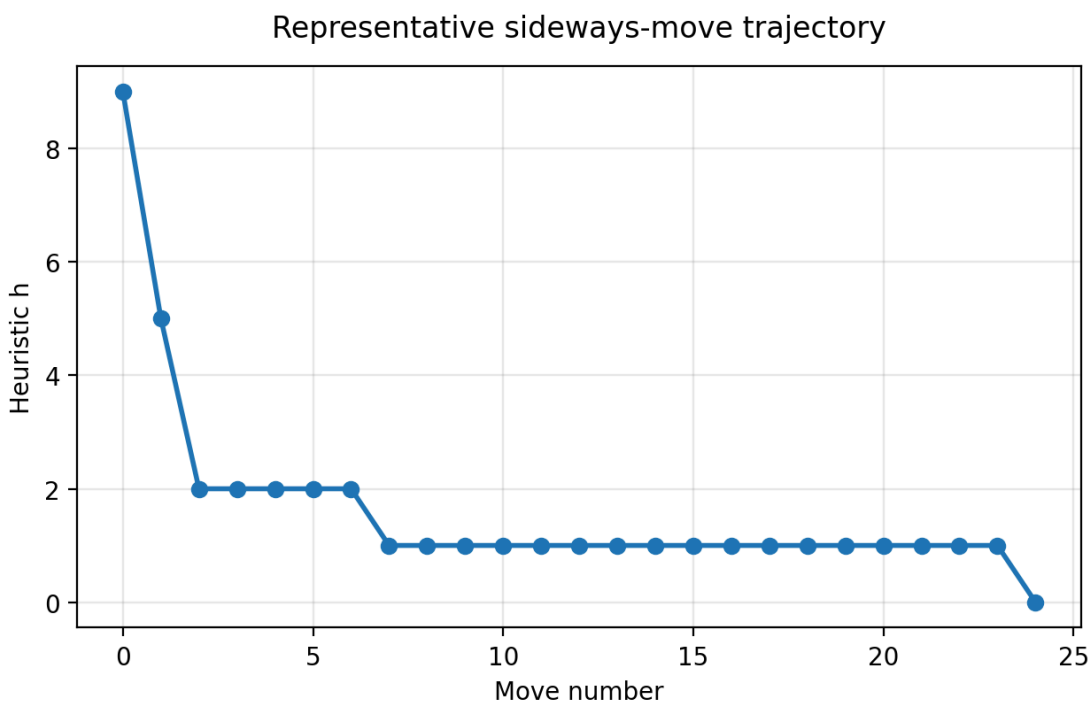

Figure 7. Representative sideways-move trajectory from Experiment Log 2.

7.1 Sequence-level observations

The basic examples show the characteristic behavior of steepest-ascent local search. The heuristic often drops quickly, but the climb can still stop with h = 1 or h = 2 because all neighboring boards are equal or worse. Low heuristic values therefore do not guarantee that h = 0 is reachable without an escape mechanism.

The sideways traces show the opposite pattern. Several sequences spend repeated moves on boards with the same heuristic value before eventually finding a strict improvement. That repeated plateau behavior is exactly why the sideways variant is both more reliable and longer on average.

8. RAW RESULTS BY EXPERIMENT LOG

The tables in this section preserve the individual statistics from each of the three saved experiment logs. Reporting the raw batches makes the stochastic variability visible and also keeps the report aligned with the exact values produced by the original program.

8.1 Experiment Log 1

Runs	Basic success %	Basic failure %	Sideways success %	Sideways failure %
50	8.00	92.00	100.00	0.00
100	11.00	89.00	91.00	9.00
200	14.00	86.00	95.50	4.50
500	12.60	87.40	92.20	7.80
1000	13.20	86.80	93.30	6.70
1500	14.73	85.27	93.93	6.07

Table 6. Success and failure rates for Experiment Log 1 (test.txt).

Runs	Basic avg. steps (success)	Basic avg. steps (failure)	Sideways avg. steps (success)	Sideways avg. steps (failure)
50	3.50	3.20	18.88	0.00
100	3.82	3.08	20.70	39.33
200	4.04	2.99	22.01	59.89
500	4.00	3.08	17.87	83.92
1000	4.07	3.06	18.94	65.72
1500	4.06	3.05	20.03	65.27

Table 7. Average step counts for Experiment Log 1 (test.txt).

Random-restart summary for Experiment Log 1 (test.txt): average restarts without sideways moves = 5.62; average queen moves without sideways moves = 21.27; average restarts with sideways moves = 0.06; average queen moves with sideways moves = 21.32.

8.2 Experiment Log 2

Runs	Basic success %	Basic failure %	Sideways success %	Sideways failure %
50	20.00	80.00	90.00	10.00
100	18.00	82.00	94.00	6.00
200	16.00	84.00	96.50	3.50
500	15.40	84.60	95.20	4.80

1000	11.70	88.30	94.60	5.40
1500	13.87	86.13	93.80	6.20

Table 8. Success and failure rates for Experiment Log 2 (test2.txt).

Runs	Basic avg. steps (success)	Basic avg. steps (failure)	Sideways avg. steps (success)	Sideways avg. steps (failure)
50	3.70	3.02	18.91	64.60
100	4.39	3.06	16.56	72.17
200	4.03	3.08	19.43	33.14
500	4.10	2.97	17.37	37.08
1000	3.99	2.99	18.49	64.17
1500	4.10	3.04	19.44	54.35

Table 9. Average step counts for Experiment Log 2 (test2.txt).

Random-restart summary for Experiment Log 2 (test2.txt): average restarts without sideways moves = 5.37; average queen moves without sideways moves = 20.44; average restarts with sideways moves = 0.08; average queen moves with sideways moves = 23.78.

8.3 Experiment Log 3

Runs	Basic success %	Basic failure %	Sideways success %	Sideways failure %
50	16.00	84.00	94.00	6.00
100	13.00	87.00	92.00	8.00
200	15.00	85.00	96.00	4.00
500	13.00	87.00	95.00	5.00
1000	13.00	87.00	95.30	4.70
1500	13.13	86.87	94.60	5.40

Table 10. Success and failure rates for Experiment Log 3 (test3.txt).

Runs	Basic avg. steps (success)	Basic avg. steps (failure)	Sideways avg. steps (success)	Sideways avg. steps (failure)
50	3.88	2.93	22.04	72.33
100	4.00	2.98	18.55	67.25
200	4.07	2.99	16.41	42.50
500	4.03	3.03	19.62	56.00
1000	3.92	3.02	19.23	57.55

1500	4.09	3.04	18.93	65.46

Table 11. Average step counts for Experiment Log 3 (test3.txt).

Random-restart summary for Experiment Log 3 (test3.txt): average restarts without sideways moves = 5.89; average queen moves without sideways moves = 22.27; average restarts with sideways moves = 0.07; average queen moves with sideways moves = 25.23.

9. CONCLUSION

The submitted implementation satisfies the core project requirements and empirically demonstrates the main strengths and weaknesses of hill climbing on the 8-Queens landscape. Plain hill climbing is fast but unreliable because it stops at local minima and plateaus. Sideways moves dramatically improve reliability by letting the search traverse flat regions, although they also increase the length of individual trajectories.

Random restart makes the overall approach practical because failed local searches can be discarded quickly and replaced with fresh random states. In the saved experiments, random restart with sideways moves was the strongest overall strategy: it almost eliminated the need to restart while still solving the problem consistently.

The most important conclusion from this project is not simply that one method has a higher success rate than another. The deeper lesson is that the structure of the search landscape determines which control strategy is effective. On the 8-Queens problem, the heuristic landscape contains many near-solutions separated by plateaus. A strategy that can move across those plateaus is therefore much more reliable than one that insists on strict improvement at every step.

Appendix A. Basic Hill-Climbing Console Traces (Experiment Log 2)

The screenshots below reproduce the full 8×8 board output for the four basic hill-climbing traces used in Section 6. Each screenshot preserves the move number, board state, and heuristic value exactly as traced from the saved experiment log.

Example 1 — Outcome: Stuck — Moves used: 4

Console Trace

```
Basic Hill Climbing | Example 1
Start board: [4, 2, 2, 4, 7, 2, 0, 0]
Outcome: Stuck | Moves used: 4
Showing steps 0 to 4

Move 0 | h=10
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | . . Q . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . Q . . . |
 4 | . . . . . . . Q |
 5 | . . Q . . . . . |
 6 | Q . . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 1 | h=6
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | . . Q . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . . Q . . |
 4 | . . . . . . . Q |
 5 | . . Q . . . . . |
 6 | Q . . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 2 | h=4
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | . . Q . . . . . |
 2 | . . . . . . . Q |
 3 | . . . . . Q . . |
 4 | . . . . . . . Q |
 5 | . . Q . . . . . |
 6 | Q . . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 3 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | . . Q . . . . . |
 2 | . . . . . . . Q |
 3 | . . . . . Q . . |
 4 | . . . . . . . Q |
 5 | . . Q . . . . . |
 6 | Q . . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+

Move 4 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | . Q . . . . . . |
 2 | . . . . . . . Q |
 3 | . . . . . Q . . |
 4 | . . . . . . . Q |
 5 | . . Q . . . . . |
 6 | Q . . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+
```

Example 2 — Outcome: Stuck — Moves used: 3

```
Console Trace

Basic Hill Climbing | Example 2
Start board: [3, 1, 3, 4, 5, 7, 7, 1]
Outcome: Stuck | Moves used: 3
Showing steps 0 to 3

Move 0 | h=9
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . Q . . . . . . |
 2 | . . . Q . . . . |
 3 | . . . . Q . . . |
 4 | . . . . . Q . . |
 5 | . . . . . . . Q |
 6 | . . . . . . . Q |
 7 | . Q . . . . . . |
   +-----------------+

Move 1 | h=6
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . Q . . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . Q . . . |
 4 | . . . . . Q . . |
 5 | . . . . . . . Q |
 6 | . . . . . . . Q |
 7 | . Q . . . . . . |
   +-----------------+

Move 2 | h=4
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . Q . . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . Q . . . |
 4 | . . . . . Q . . |
 5 | . . . . . . . Q |
 6 | . . . . Q . . . |
 7 | . Q . . . . . . |
   +-----------------+

Move 3 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | Q . . . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . Q . . . |
 4 | . . . . . Q . . |
 5 | . . . . . . . Q |
 6 | . . . . Q . . . |
 7 | . Q . . . . . . |
   +-----------------+
```

Example 3 — Outcome: Solved — Moves used: 3

Console Trace

```
Basic Hill Climbing | Example 3
Start board: [6, 5, 5, 3, 4, 3, 7, 4]
Outcome: Solved | Moves used: 3
Showing steps 0 to 3

Move 0 | h=8
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . . . . Q . . |
 2 | . . . . . Q . . |
 3 | . . . Q . . . . |
 4 | . . . . Q . . . |
 5 | . . . Q . . . . |
 6 | . . . . . . . Q |
 7 | . . . . Q . . . |
   +-----------------+

Move 1 | h=4
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . . . . Q . . |
 2 | . . . . . Q . . |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . Q . . . . |
 6 | . . . . . . . Q |
 7 | . . . . Q . . . |
   +-----------------+

Move 2 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . . . . Q . . |
 2 | . . . . . Q . . |
 3 | . . Q . . . . . |
 4 | Q . . . . . . . |
 5 | . . . Q . . . . |
 6 | . . . . . . . Q |
 7 | . . . . Q . . . |
   +-----------------+

Move 3 | h=0
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . Q . . . . . . |
 2 | . . . . . Q . . |
 3 | . . Q . . . . . |
 4 | Q . . . . . . . |
 5 | . . . Q . . . . |
 6 | . . . . . . . Q |
 7 | . . . . Q . . . |
   +-----------------+
```

Example 4 — Outcome: Solved — Moves used: 4

Console Trace

```
Basic Hill Climbing | Example 4
Start board: [6, 2, 5, 6, 7, 0, 1, 3]
Outcome: Solved | Moves used: 4
Showing steps 0 to 4

Move 0 | h=6
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . Q . . . . . |
 2 | . . . . . Q . . |
 3 | . . . . . . Q . |
 4 | . . . . . . . Q |
 5 | Q . . . . . . . |
 6 | . Q . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+

Move 1 | h=4
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . Q . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . . . Q . |
 4 | . . . . . . . Q |
 5 | Q . . . . . . . |
 6 | . Q . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+

Move 2 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . Q . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . . Q . . |
 4 | . . . . . . . Q |
 5 | Q . . . . . . . |
 6 | . Q . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+

Move 3 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . Q . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . . Q . . |
 4 | . . . . . . . Q |
 5 | . . . . Q . . . |
 6 | . Q . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+

Move 4 | h=0
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . Q . . . . . |
 2 | Q . . . . . . . |
 3 | . . . . . Q . . |
 4 | . . . . . . . Q |
 5 | . . . . Q . . . |
 6 | . Q . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+
```

Appendix B. Sideways-Move Console Traces (Experiment Log 2)

The screenshots below reproduce the full 8×8 board output for the four sideways-move traces used in Section 6. Longer traces are split across multiple consecutive screenshots so that the boards remain legible.

Example 1 — Outcome: Solved — Moves used: 24

```
Console Trace

Hill Climbing with Sideways Moves | Example 1
Start board: [3, 5, 0, 0, 0, 1, 7, 0]
Outcome: Solved | Moves used: 24
Showing steps 0 to 4

Move 0 | h=9
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | Q . . . . . . . |
 3 | Q . . . . . . . |
 4 | Q . . . . . . . |
 5 | . Q . . . . . . |
 6 | . . . . . . . Q |
 7 | Q . . . . . . . |
   +-----------------+

Move 1 | h=5
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | Q . . . . . . . |
 3 | Q . . . . . . . |
 4 | . . . . Q . . . |
 5 | . Q . . . . . . |
 6 | . . . . . . . Q |
 7 | Q . . . . . . . |
   +-----------------+

Move 2 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | Q . . . . . . . |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . Q . . . . . . |
 6 | . . . . . . . Q |
 7 | Q . . . . . . . |
   +-----------------+

Move 3 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | Q . . . . . . . |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . . . . . . . Q |
 7 | Q . . . . . . . |
   +-----------------+

Move 4 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | Q . . . . . . . |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . . . . . . . Q |
 7 | . Q . . . . . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 1
Start board: [3, 5, 0, 0, 0, 1, 7, 0]
Outcome: Solved | Moves used: 24
Showing steps 5 to 9

Move 5 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | Q . . . . . . . |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . . . . . . . Q |
 7 | Q . . . . . . . |
   +-----------------+

Move 6 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . . . . . . . Q |
 7 | Q . . . . . . . |
   +-----------------+

Move 7 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 8 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 9 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 1
Start board: [3, 5, 0, 0, 0, 1, 7, 0]
Outcome: Solved | Moves used: 24
Showing steps 10 to 14

Move 10 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 11 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . . . . Q . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 12 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 13 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 14 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . Q . . . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 1
Start board: [3, 5, 0, 0, 0, 1, 7, 0]
Outcome: Solved | Moves used: 24
Showing steps 15 to 19

Move 15 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . . Q . . . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 16 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . Q . . . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 17 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . Q . . . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 18 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | Q . . . . . . . |
 2 | . . . . . . . Q |
 3 | . . Q . . . . . |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 19 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | Q . . . . . . . |
 2 | . . . . . . . Q |
 3 | . . . . . . . Q |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 1
Start board: [3, 5, 0, 0, 0, 1, 7, 0]
Outcome: Solved | Moves used: 24
Showing steps 20 to 24

Move 20 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | Q . . . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . . . . Q |
 4 | . . . . Q . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 21 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | Q . . . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . . . . Q |
 4 | . Q . . . . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 22 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | Q . . . . . . . |
 2 | . . Q . . . . . |
 3 | . . . . . . . Q |
 4 | . Q . . . . . . |
 5 | . . . . . . Q . |
 6 | Q . . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 23 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . . Q . |
 2 | . . Q . . . . . |
 3 | . . . . . . . Q |
 4 | . Q . . . . . . |
 5 | . . . . . . Q . |
 6 | Q . . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+

Move 24 | h=0
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . . Q . |
 2 | . . Q . . . . . |
 3 | . . . . . . . Q |
 4 | . Q . . . . . . |
 5 | . . . . Q . . . |
 6 | Q . . . . . . . |
 7 | . . . . . Q . . |
   +-----------------+
```

Example 2 — Outcome: Solved — Moves used: 20

```
Console Trace

Hill Climbing with Sideways Moves | Example 2
Start board: [6, 5, 4, 1, 7, 4, 0, 0]
Outcome: Solved | Moves used: 20
Showing steps 0 to 4

Move 0 | h=8
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . . . . Q . . |
 2 | . . . . Q . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . . Q |
 5 | . . . . Q . . . |
 6 | Q . . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 1 | h=4
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . . Q |
 5 | . . . . Q . . . |
 6 | Q . . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 2 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . Q . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . . Q |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 3 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . . Q |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 4 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 2
Start board: [6, 5, 4, 1, 7, 4, 0, 0]
Outcome: Solved | Moves used: 20
Showing steps 5 to 9

Move 5 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . Q . . . |
   +-----------------+

Move 6 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . . Q |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . Q . . . |
   +-----------------+

Move 7 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . . Q |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 8 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 9 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 2
Start board: [6, 5, 4, 1, 7, 4, 0, 0]
Outcome: Solved | Moves used: 20
Showing steps 10 to 14

Move 10 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 11 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . . Q |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 12 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 13 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | Q . . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 14 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 2
Start board: [6, 5, 4, 1, 7, 4, 0, 0]
Outcome: Solved | Moves used: 20
Showing steps 15 to 19

Move 15 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 16 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . . Q |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 17 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . . Q |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . Q . . . |
   +-----------------+

Move 18 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . . Q |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | Q . . . . . . . |
 6 | . . Q . . . . . |
 7 | . . . . Q . . . |
   +-----------------+

Move 19 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . . . . Q |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | Q . . . . . . . |
 6 | . . Q . . . . . |
 7 | . . . . Q . . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 2
Start board: [6, 5, 4, 1, 7, 4, 0, 0]
Outcome: Solved | Moves used: 20
Showing steps 20 to 20

Move 20 | h=0
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . Q . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . . . Q |
 3 | . Q . . . . . . |
 4 | . . . . . . Q . |
 5 | Q . . . . . . . |
 6 | . . Q . . . . . |
 7 | . . . . Q . . . |
   +-----------------+
```

Example 3 — Outcome: Solved — Moves used: 3

Console Trace

```
Hill Climbing with Sideways Moves | Example 3
Start board: [4, 0, 7, 1, 2, 6, 5, 7]
Outcome: Solved | Moves used: 3
Showing steps 0 to 3

Move 0 | h=5
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | Q . . . . . . . |
 2 | . . . . . . . Q |
 3 | . Q . . . . . . |
 4 | . . Q . . . . . |
 5 | . . . . . . Q . |
 6 | . . . . . Q . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 1 | h=4
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | Q . . . . . . . |
 2 | . . . . . . . Q |
 3 | . . . . . Q . . |
 4 | . . Q . . . . . |
 5 | . . . . . . Q . |
 6 | . . . . . Q . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 2 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | Q . . . . . . . |
 2 | . . . . . . . Q |
 3 | . . . . . Q . . |
 4 | . . Q . . . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 3 | h=0
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . . . . Q . . . |
 1 | Q . . . . . . . |
 2 | . . . . . . . Q |
 3 | . . . . . Q . . |
 4 | . . Q . . . . . |
 5 | . . . . . . Q . |
 6 | . Q . . . . . . |
 7 | . . . Q . . . . |
   +-----------------+
```

Example 4 — Outcome: Solved — Moves used: 11

Console Trace

```
Hill Climbing with Sideways Moves | Example 4
Start board: [1, 5, 4, 7, 6, 4, 0, 0]
Outcome: Solved | Moves used: 11
Showing steps 0 to 4

Move 0 | h=8
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . . Q . . . |
 3 | . . . . . . . Q |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | Q . . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 1 | h=4
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . Q . . |
 2 | . . Q . . . . . |
 3 | . . . . . . . Q |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | Q . . . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 2 | h=3
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . Q . . |
 2 | . . Q . . . . . |
 3 | . . . . . . . Q |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 3 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . Q . . |
 2 | . . Q . . . . . |
 3 | Q . . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+

Move 4 | h=2
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | Q . . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | Q . . . . . . . |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 4
Start board: [1, 5, 4, 7, 6, 4, 0, 0]
Outcome: Solved | Moves used: 11
Showing steps 5 to 9

Move 5 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . Q . . . . |
 3 | Q . . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 6 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . Q . . |
 2 | . . . . . Q . . |
 3 | Q . . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 7 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . Q . . . . |
 2 | . . . . . Q . . |
 3 | Q . . . . . . . |
 4 | . . . . . . Q . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 8 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . Q . . . . |
 2 | . . . . . Q . . |
 3 | Q . . . . . . . |
 4 | . . Q . . . . . |
 5 | . . . . Q . . . |
 6 | . . Q . . . . . |
 7 | . . . . . . . Q |
   +-----------------+

Move 9 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . Q . . . . |
 2 | . . . . . Q . . |
 3 | Q . . . . . . . |
 4 | . . Q . . . . . |
 5 | . . . . Q . . . |
 6 | . . . . . . Q . |
 7 | . . . . . . . Q |
   +-----------------+
```

Console Trace

```
Hill Climbing with Sideways Moves | Example 4
Start board: [1, 5, 4, 7, 6, 4, 0, 0]
Outcome: Solved | Moves used: 11
Showing steps 10 to 11

Move 10 | h=1
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . Q . . . . |
 2 | . . . . . Q . . |
 3 | Q . . . . . . . |
 4 | . . Q . . . . . |
 5 | . . . . Q . . . |
 6 | . . . . . . Q . |
 7 | . . . Q . . . . |
   +-----------------+

Move 11 | h=0
     0 1 2 3 4 5 6 7
   +-----------------+
 0 | . Q . . . . . . |
 1 | . . . . . . . Q |
 2 | . . . . . Q . . |
 3 | Q . . . . . . . |
 4 | . . Q . . . . . |
 5 | . . . . Q . . . |
 6 | . . . . . . Q . |
 7 | . . . Q . . . . |
   +-----------------+
```